CS2750

THE UNIVERSITY OF WARWICK

Paper Details

Paper Code: CS2750

Paper Title: Probability and Statistics

Exam Period: Summer 2025

Exam Rubric

Time allowed: 3 hours

Exam Type: Written – Restricted

Approved Calculators: CASIO FX82, FX83 and FX85 series

Additional Stationery

Instructions

Answer **FIVE** questions out of the **SIX** possible.

If more then **FIVE** questions are answered, please clearly indicate which questions you wish to be marked. Otherwise, the first **FIVE** questions answered in the answer booklet will be marked.

For questions requiring you to compute some quantities (e.g., probabilities or expectations) you must show your reasoning. There is no need to compute the numbers in decimal form, e.g., it is entirely appropriate to leave an answer like $\frac{324 \cdot 142 \cdot 84}{59523 \cdot 707} + \frac{\sqrt{275}}{294}$. In general, any arithmetic expression does not need to be computed as long as it is explicit; definite integrals, however, must be solved.

Read carefully the instructions on the answer booklet and make sure that the particulars required are entered on each answer booklet.

One sheet of paper (informally called a "crib sheet") containing handwritten notes of student's own choice can be brought and used as an individual reference during the exam. The handwriting must be of reasonable size, e.g., similar to that used in exam writing. The crib sheet must be treated as an incentive to lessen memorization and focus instead on understanding the class materials and developing problem solving skills; solely relying on the crib sheet is likely to lead to failing the exam.

The exam question paper MUST NOT be removed from the examination venue.

---

1. [20 marks]

   Thirty students go to a party and leave their coats in a messy cloakroom. When the party is over, the host returns the students' coats in a random order.

   (a) Describe the sample space Ω of the underlying random experiment. How many elements does Ω contain? [5]

   (b) What is the probability that a particular student gets the correct coat back? [5]

   (c) What is the expected number of students who get their own coats back? [10]

---

2. [20 marks] A family has two children. Each child can be either a boy or a girl with equal probabilities, and each child can be born in any of the weekdays with equal probabilities.
   (a) What is the probability of having two boys given that at least one child is a boy? [5]
   (b) What is the probability of having two boys given that the family has a boy born on a Tuesday? [10]
   (c) Compute the expected number of boys, born either on Monday or Tuesday, that the family has. [5]

---

3. [20 marks]

   In each drawing of the 6/59 lottery, six different integers from $\{1, 2, \ldots, 59\}$ are randomly chosen.

   (a) What is the expected number of integers from $\{1, 2, \ldots, 59\}$ which *do not* show up in 10 drawings? [5]

   (b) What is the standard deviation for the same quantity from (a), i.e., the number of integers from $\{1, 2, \ldots, 59\}$ which *do not* show up in 10 drawings? [15]

---

4. [20 marks]

   Let X and Y be random variables such that $X \perp Y$ (i.e., they are independent).

   (a) Describe $\mathrm{Var}[X - 100]$ in terms of $\mathrm{Var}[X]$. Prove it! [5]

   (b) Describe $\mathrm{Var}[X - Y]$ in terms of $\mathrm{Var}[X]$ and $\mathrm{Var}[Y]$. Prove it! [10]

   (c) Prove whether $\max(X, Y) \perp X$ or not. [5]

---

5. [20 marks]

   (a) Explain how the exponential distribution can be obtained from the geometric distributions through a limiting argument. [5]

   (b) Consider a Poisson arrival process $N(t)$ with parameter λ. What is the distribution of the time of the first arrival conditioned on the event that there is exactly one arrival by time 10? [5]

   (c) Consider that, at time 0, you are at a bus stop where buses arrive as a Poisson (arrival) process with parameter λ; once a bus arrives, the time it would take you to get on the bus is exponentially distributed with parameter μ. What is the probability that you are in the bus by time t conditioned on the event that there has been a single arrival by time t? [10]

---

6. [20 marks] Thirty students are each playing some individual game, each day, and independently; the probability of winning in each game is p. We sample X_i as the number of days it takes for student i to first win, for $i = 1, 2, \ldots, 30$.

   (a) Derive the ML estimator $\hat{p}(X_1, X_2, \ldots, X_{30})$ for p. [10]

   (b) Is your estimator from (a) unbiased? Justify! [10]
